THE QUEST

Health tracking is easy to start and hard to keep up. In HealthWiz Kingdom every healthy action has a visible effect.

It is a pixel-art fantasy RPG that is also a health tracker. You log **water, meals, sleep, stairs** (with workout heart rate), **runs** and **stress**. **Medius, the Wizard King,** guides you as your real actions restore the kingdom, with **Luna,** an apprentice witch, as your everyday companion.







REAL SCREENS FROM THE APP (SAMPLE DATA)

HOW IT WORKS

1 LOG
a real habit: a glass of water, a meal, a climb

2 RESTORE
regions of the kingdom heal as you keep going

3 UNDERSTAND
a "?" shows the formula and sources behind every number

WHAT SETS IT APART

1. **Health first, game second.** Health numbers and units stay in a plain, readable face. Game values (XP, levels, badges) are tagged "game progress, not a health measurement".
2. **Every number shows its working.** Inputs, formula, your own numbers step by step, linked sources and limits.
3. **Honest about uncertainty.** Each value is labelled **measured, calculated, estimated, simulated** or **missing**. No invented readings, no made-up error margins.

8	8	104	102
KINGDOM REGIONS	COLOUR THEMES	CODE MODULES	TEST FILES

3 UN SUSTAINABLE DEVELOPMENT GOAL 3
Good Health and Well-being

BEGIN YOUR QUEST

1. **Scan** or open the link. It runs in any modern browser; nothing to install.
2. Create your traveller in the **Traveller's Registry**.
3. Log one thing today. Medius will take it from there.
4. Tap **Install app** or **Add to Home Screen** to play offline.

healthwizkingom.github.io/Healthwiz-Kingdom

YOUR DATA STAYS YOURS

- Saved **on your device**. Download or restore a backup any time.
- Free account is **optional**: cloud backup and sync. Row Level Security means only you can read your row.
- **Delete account and cloud data** in two taps.
- Location is read **only when you tap**, and is never saved or sent. Camera pulse video is never stored.
- Medius AI chat is **never stored**, and there is no AI key inside the app.

UNDER THE HOOD

A static web app: plain JavaScript, no build step, no server code of its own. Installable and offline-ready.

PWA + service worker · Supabase · Postgres + RLS · Edge Functions · Google Gemini · MapLibre GL · Three.js · WebXR · Web Bluetooth · Open-Meteo · Blender art

CREDITS AND LICENCES

Fonts: Press Start 2P and Pixelify Sans (SIL OFL 1.1). Map data © OpenStreetMap contributors (ODbL). MapLibre GL JS (BSD-3-Clause). Three.js (MIT). Map avatars: Tom, the anime school boy and girl (CC BY 4.0, suzuart). Campus look follows a drone video by @ikraMRahim, used with the creator's permission. Knight 3D base: MakeHuman CC0 assets. Every other character and scene is an original HealthWiz design.

Wellness tracker, not a medical device. HealthWiz does not diagnose anything and gives no medical advice. If you feel unwell, talk to a parent, teacher or healthcare professional. In an emergency call **999** (Malaysia).



Turn your health into an adventure.

A pixel-art fantasy RPG that is also a health tracker, built on real science and honest numbers.

GROUP 14
MARA · Kolej MARA Kulim

RAYYAN RAFIQIN	QAYYUM	NAUFAL
AMIR	MUI'ZZUDDIN	MUIZ ZIKRI

Wellness tracker, not a medical device



TRACK

THE HEALTH HALL · EIGHT TRACKERS IN ONE PLACE



PLAY & GUIDE

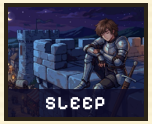
YOUR LOGS RESTORE THE KINGDOM



PROVE IT

SCIENCE YOU CAN CHECK









WATER QUEST Your daily goal comes from your weight (35 mL/kg: the team's own rule, and the "?" says so). The knight walks, drinks, carries and pours as you log. Opt in to live weather and the Elemental Thermometer adjusts your water potion.

NUTRITION Log from the Kolej MARA Kulim canteen menu with calories. Signed in, **Luke** searches the web for any food and portion, shows a source for every part and checks the energy with 4·protein + 4·carb + 9·fat. All estimates are labelled.

SLEEP Hours against National Sleep Foundation ranges, a Sleep Score (our own game formula, labelled as such), the Dream Battle, Counting Sheep and a wake-up alarm for Android, iPhone and calendar.

STRESS QUEST The DASS-21 check-in (21 statements, three scales, no severity labels), the Storm Within, Calming Grove breathing and five relief activities. Crisis help numbers are always on screen.

BODY & ENERGY BMI (an adult screening number, not a diagnosis), BMR by Mifflin–St Jeor, daily energy = BMR × FAO/WHO/UNU activity level, plus protein, carb, fat and fibre goals.

TRAINING HALL Goal-based sessions with 22 moves and 19 muscle maps; Calisthenics is bodyweight only. "I feel dizzy" and "Chest pain" stop buttons are always at hand.

RUNNING ROAD GPS run tracker: distance, moving time, pace and calories (MET × kg × h, 2024 Compendium), with an opt-in Runners' Board.

STAIRS & HEART RATE Log BPM before, after and one minute after a climb for recovery. Measure pulse with your phone camera (the video is never stored) or pair a smartwatch or chest strap over Web Bluetooth.

EVERYWHERE Log for yesterday or any earlier date, browse every entry by day with totals, and print a weekly report.



MEDIUS
The Wizard King. Tutorials, insights and the Wizard's Counsel.



LUNA
Apprentice witch. Everyday companion with 21 poses.



LUKE
Wannabe wizard. Finds foods and teaches the lab.

THE LIVING KINGDOM Eight regions (Dream Realm, Stair Mountain, Energy Forge, Mind Forest, Heartstone Hall, Water Valley, Nutrition Village, Balance Tower), each in one of five states up to **Flourishing**, set by the days you logged this week.

QUESTS & REWARDS An adaptive daily focus quest and weekly quests. Fair XP with a daily logging allowance, levels, audited badges and gentle streaks. Mini-games (Dream Battle, Storm Within, Calming Grove, Counting Sheep) turn each log into a short scene.

HEALTH SCORE, 0-100 Five weighted indicators plus fruit and veg servings. In the **Health Orb** Medius reads your score and turns it into ranked quest cards: Start Here, Next Step, Keep Going.

MEDIUS NOTICES Insights from your own baselines, a what-if simulator, week-against-week stats and the **Health Time Machine** timeline.

WIZARD'S COUNSEL Talk with Medius or Luna (Google Gemini, through a secure server function). Signed-in only; Medius is limited to 30 messages an hour, with safety rules, a gentle fallback and crisis numbers. Nothing is stored. **Ask Luna** finds any feature without AI.

ARCANA UPGRADES The Oracle's crystal ball (readiness from sleep, DASS-21 and recovery), the Guild War of Joules and the Dueling Arena: race a replay of your own earlier climb.

OPT-IN COMMUNITY The Hall of Heroes shows game progress only; the Runners' Board shows a nickname and weekly totals only.

MADE FOR EVERYONE Eight themes, performance modes, reduced motion, keyboard and screen-reader friendly charts.

CAMPUS HEALTH OBSERVATORY A 3D map of Kolej MARA Kulim built from OpenStreetMap, in Realistic and Fantasy looks. Every staircase is a spinning heart coloured Mild, Moderate or Vigorous: tap one to log a climb or workout. GPS check-in finds the nearest stairs; your position is never saved or sent.

STAIR SENSE & STAIR SIGHT Height, work and power from the stairs:

$h = n \times r$ $W = m \cdot g \cdot h$ $P = W \div t$

Camera edge-finding, and WebXR on Android, measure a real staircase's rise and tread.

THE GROUP 14 STAIR EXPERIMENT

Six participants climbed measured stairways at different intensities. The pulse is modelled as **periodic motion** (frequency, period, angular frequency) and VO_2 is estimated with the project's own method. Automated tests **recompute the report's tables from the raw counts**. If the pulse after is lower than before, the app shows "not valid" instead of a number.

EVIDENCE VAULT Every source on a shelf: source, claim, how HealthWiz uses it and where you see it. Each "?" links straight to its entry.

SCIENCE LABORATORY **Be Your Own Experiment**, the **Evidence Trial** (reasoning questions built from real data) and the **Uncertainty Engine**, which shows how unmeasured assumptions move an estimate.

ASCENT MONUMENT Your stair climbing builds an endless Tower of Babel, one stone for every 0.25 m, compared with real landmarks.

SAFE DATA Saved data is versioned and upgraded step by step; a backup copy is made before any upgrade or repair.

Sources behind the numbers: Mifflin–St Jeor (energy) · FAO/WHO/UNU (activity levels) · 2024 Compendium of Physical Activities (MET) · Keytel (heart-rate energy) · Tanaka (peak heart rate) · Lovibond & Lovibond (DASS-21) · National Sleep Foundation · WHO (fruit and veg, diet) · IOM (macros). Where a rule is the team's own, the "?" says so.